

How Predictable Is ASML?

The honest deliverable is not a forecast. It is a **verdict on forecastability**.

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github.com/AdityaMehta2006/ts_mini_project

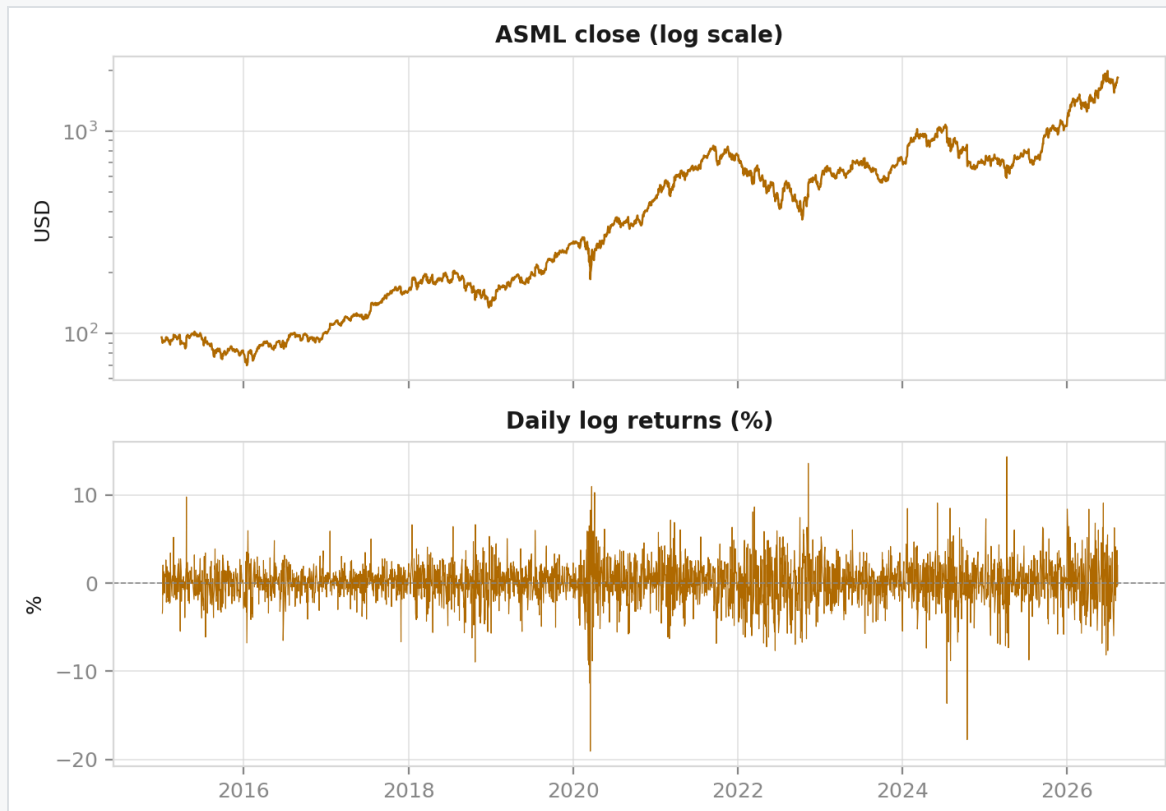
The data

PROPERTY	VALUE
Ticker	ASML
Source	Yahoo Finance, adjusted
Period	2015-01-02 → 2026-08-14
Daily observations	2,921
Monthly observations	140

PROPERTY	VALUE
First close	95.94
Last close	1,844.08
Total return	1,822.0%
CAGR	28.98%
Annualised volatility	38.01%

The EUV lithography monopoly — structurally important, and violently volatile. Daily data for stationarity, ARIMA and volatility; monthly for decomposition and smoothing.

Why we model the log price



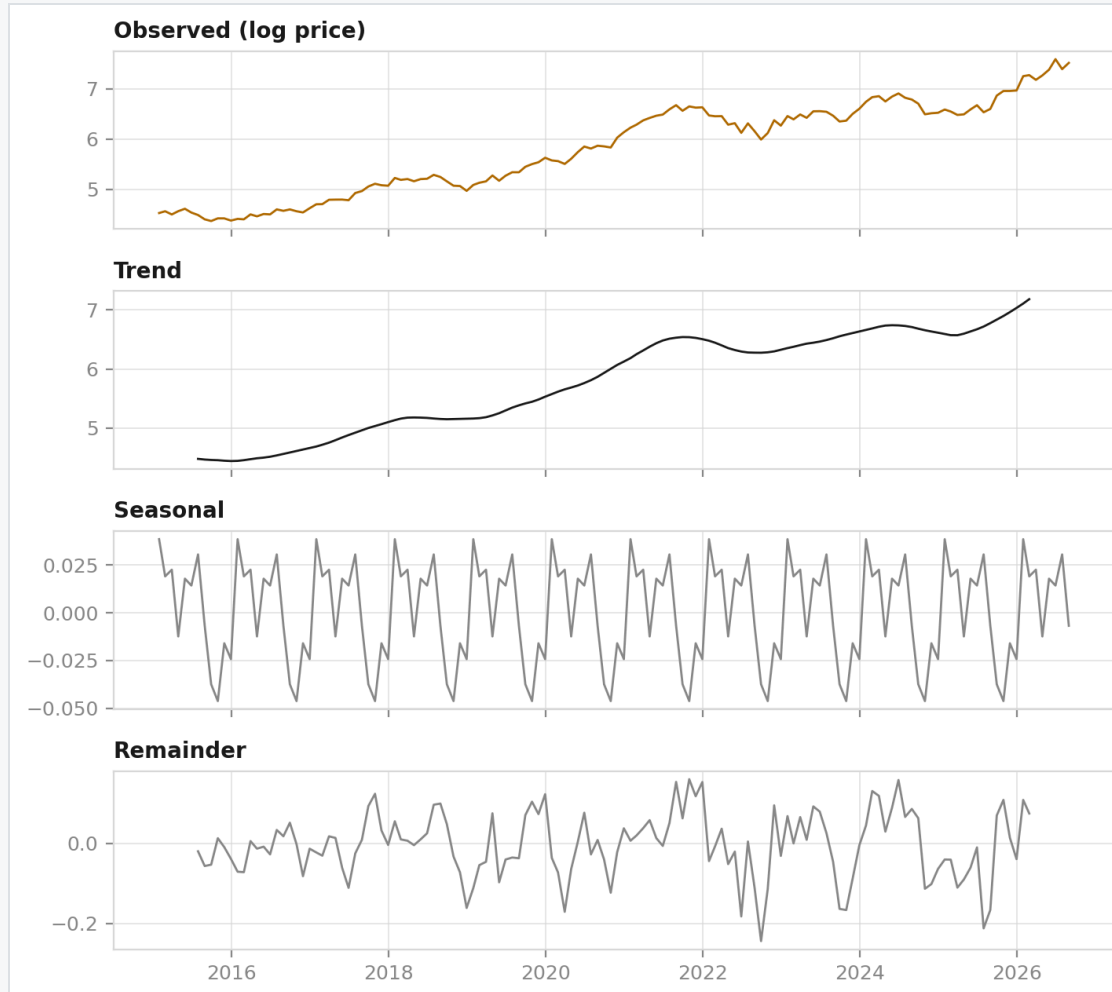
ASML compounded at 28.98% a year.

Growth that fast is **multiplicative** — variance grows with the level.

Logs turn multiplicative growth into additive growth *and* stabilise variance in one step.

Returns: skew -0.33 , kurtosis 7.81 (normal = 3). Those fat tails come back twice.

There is no calendar season in a share price



COMPONENT	STRENGTH
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Trend	0.99
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Seasonal	0.10
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A seasonal strength of 0.10 means that panel is fitting noise.

A negative result — and it makes a prediction: Holt-Winters' seasonal term should earn nothing. Part 2 confirms $\gamma = 0$ exactly.

Stationarity: two tests, opposite nulls



ADF H_0 : unit root • KPSS H_0 : stationary

SERIES	ADF P	KPSS P
Log price	0.9634	0.0100
Δ log price	< 0.0001	0.1000

Log price: all 40 lags outside the ± 0.0363 band.

Returns: only 9 of 40, none large.

Ljung-Box on returns gives $p < 0.0001$, so returns are not *strictly* white noise. At $n = 2,920$ even $r = 0.04$ is detectable. Real, and far too small to trade.

Because the nulls are opposed, agreement in both directions is far stronger than either test alone — ASML log price is $I(1)$, so $d = 1$

The three-model ladder

SES → level only

Holt → + trend

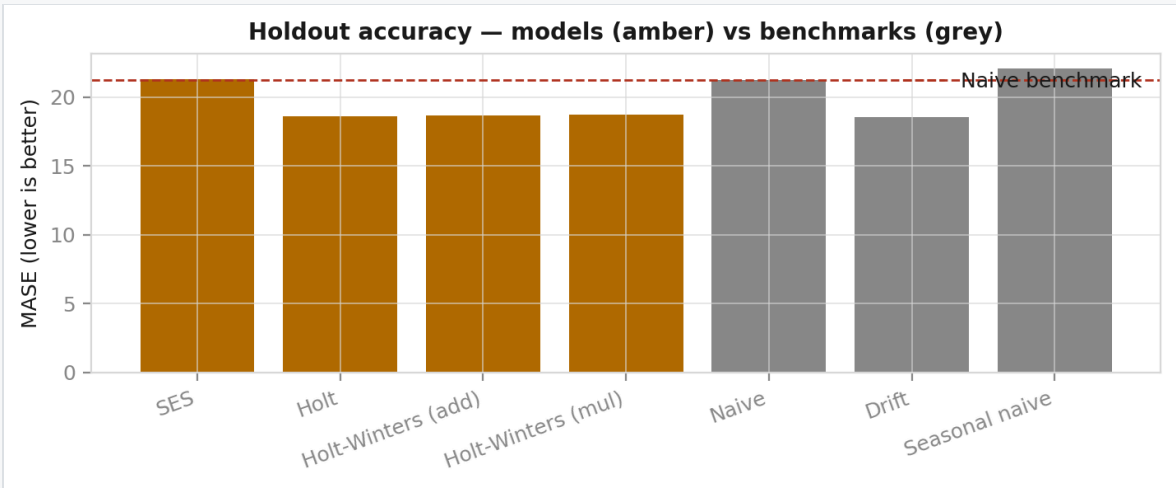
Holt-Winters → + season

MODEL	A	B	Γ	MAPE
SES	0.9879	—	—	44.68%
Holt	0.9371	0.0000	—	39.19%
Holt-Winters	1.0000	0.0000	0.0000	39.60%

Three parameters tell the whole story:

- $\alpha \approx 1$ → the level is reset to whatever just happened — *a random walk in a costume*
- $\beta = 0$ → trend treated as fixed, so Holt collapses onto Drift
- $\gamma = 0$ → no seasonality to learn, exactly as Part 1 predicted

Nothing beats Drift

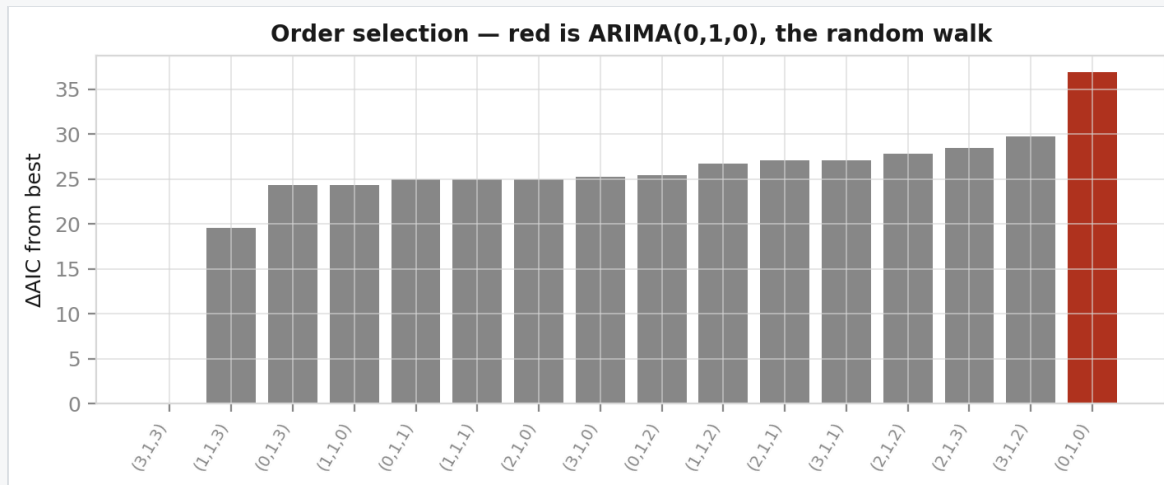


MODEL	MASE	BENCHMARK	MASE
SES	21.279	Naive	21.248
Holt	18.625	Drift	18.565
Holt-Winters	18.653	Seasonal naive	22.081

Last value plus the average monthly change — two parameters, no fitting — wins.

MASE divides by the one-step in-sample naive error, so values above 1 are normal at a 12-step horizon. Read the ranking, not the level.

Order selection: 16 candidates by AIC



ORDER	AIC	ΔAIC
(3,1,3)	-13,539.73	0.00
(1,1,3)	-13,520.20	19.53
(0,1,0) random walk	-13,502.80	36.93

The random walk ranks 16th of 16.

AIC *does* find real structure in the differenced series — this is not a criterion unable to choose.

But the coefficients mislead

COEFFICIENT	ESTIMATE
ar.L1	-0.7897
ar.L2	0.8681
ar.L3	0.9119
ma.L1	0.7384
ma.L2	-0.8740
ma.L3	-0.8520

All six significant at 5%. It looks like strong structure.

A model can carry large, significant coefficients and still be a near-random walk. Only an out-of-sample test settles it.

SMALLEST ROOT MODULUS	
AR(3)	1.0026
MA(3)	1.0035

Both sit almost exactly *on* the unit circle, and almost exactly on top of each other — the polynomials share a **near-common factor** and very nearly cancel.

A factor on both sides of the equation does no forecasting work at all.

The trap: one residual inverts every diagnostic

`statsmodels` starts its state-space filter from a diffuse prior. The first residual is the filter finding its footing — **not** a model error.

192.2 σ

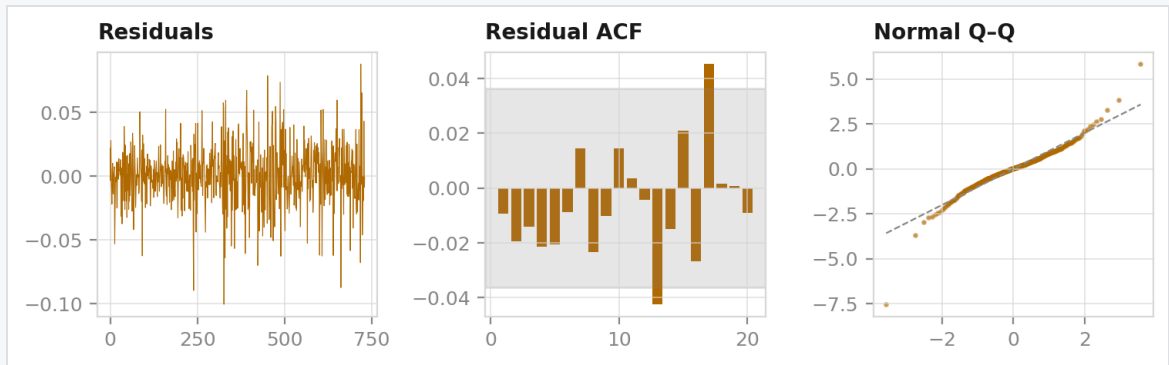
FIRST RESIDUAL 4.5638 VS $\Sigma = 0.0237$

LJUNG - BOX	P
Burn-in kept	0.99999
Burn-in trimmed	0.0961

Kept, it dominates every sum of squares that follows and the model looks like flawless white noise. Trimmed, it passes — but only just.

The whole appearance of a perfect fit rested on one observation, and nothing in the output looks wrong.

Residuals, and where the model genuinely fails



TEST	P	MEANING
Ljung-Box (10)	0.0961	no autocorrelation
Jarque-Bera	< 0.0001	not normal
ARCH-LM (10)	< 0.0001	variance not constant

HORIZON	COVERAGE	NOMINAL
h = 1	82.4%	95%
h = 3	82.4%	95%
h = 6	79.7%	95%

Mean 81.5% against 95% — the intervals are **too narrow**. ARIMA assumes one constant variance; ARCH-LM rejects that at $p < 0.0001$.

No model beats naive at any horizon

MODEL	H=1	H=3	H=6
ARIMA(3,1,3)	7.179	13.558	24.483
Naive	7.151	13.634	24.556
Drift	7.020	13.211	23.273

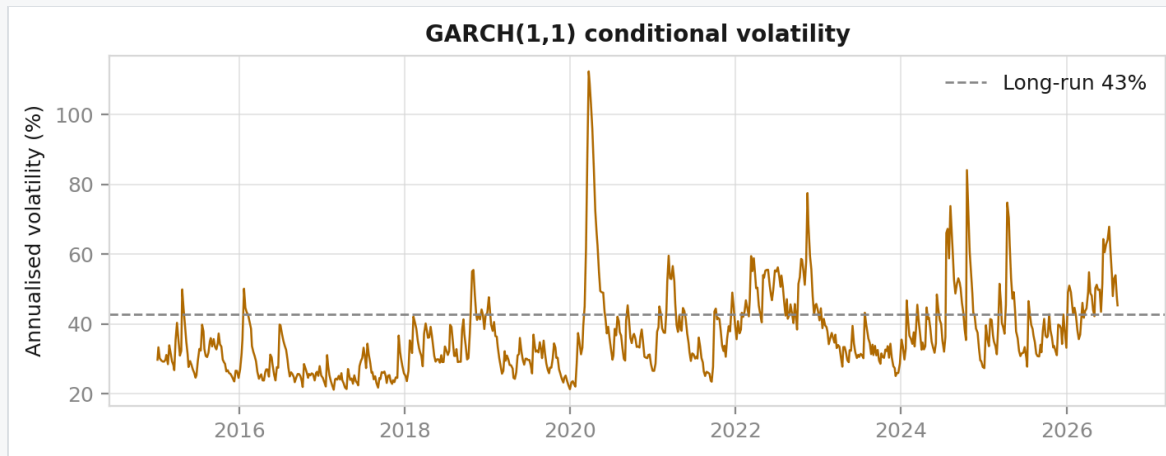
74 rolling origins, expanding window, refit every time. MASE by horizon.

Diebold–Mariano p vs naive: 0.235 → 0.888.

Not one below 0.05.

Drift ranks first at every horizon — but a ranking without a significant margin is **not a win**, and reporting it as one would be the easiest mistake in this project.

The variance, however, is forecastable



$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

PARAMETER	VALUE
α (shock)	0.0752
β (persistence)	0.9113
$\alpha + \beta$	0.9864
Half-life	50.7 days

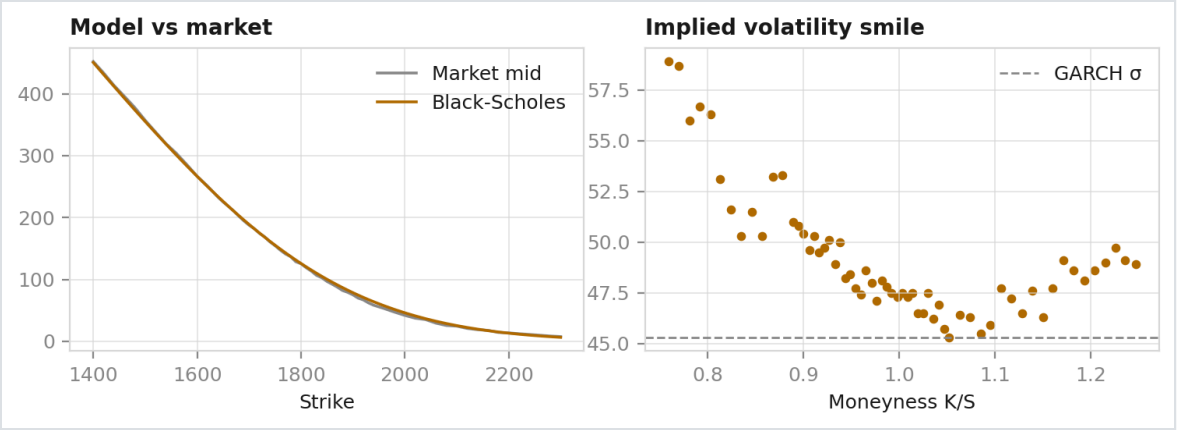
Ljung-Box on squared standardised residuals: $p = 0.2012 \rightarrow$ ARCH fully absorbed.

We cannot say where ASML is going. We can say how rough the ride will be.

- **Mean** — unforecastable. 74 origins, no significant edge over naive.
- **Variance** — strongly forecastable. Persistence 0.9864, half-life 50.7 days.

The same data supports both claims, and they are not in tension: returns are unpredictable in *direction* and highly structured in *magnitude*.

Black–Scholes against a live market

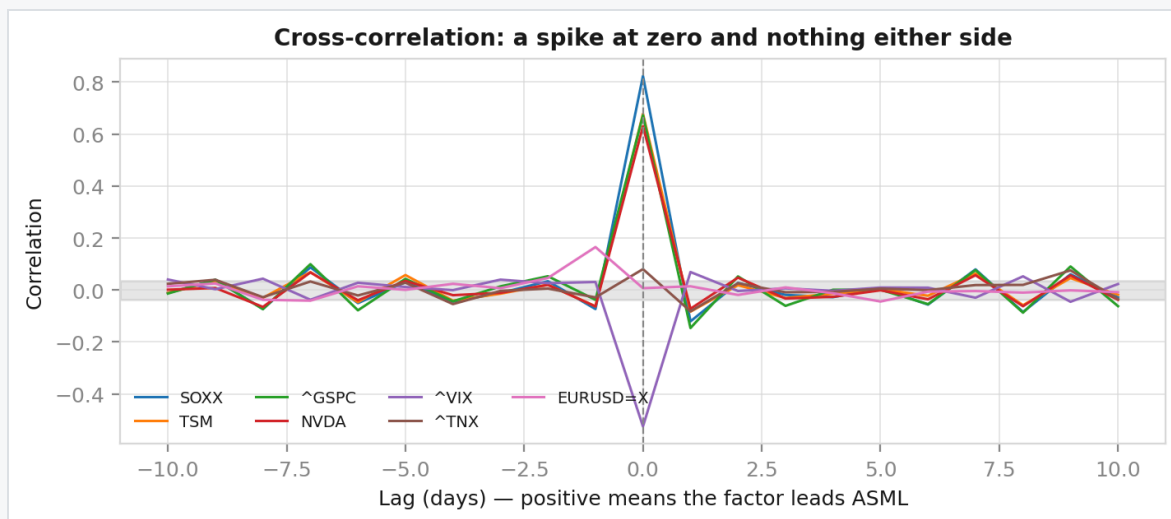


61 near-the-money calls, expiry 2026-09-18, $r = 3.697\%$

Spot	1,844.08
ATM strike	1,840
Market mid	102.85
Black-Scholes @ GARCH σ	105.09
Implied vol	49.2%
GARCH vol	45.3%
Gap	+3.9 pts

Black–Scholes assumes **one volatility for every strike**; implied vol visibly **curves**. Both departures trace to the constant-Gaussian assumption Part 3 had already rejected.

If ASML can't predict itself, can anything else?

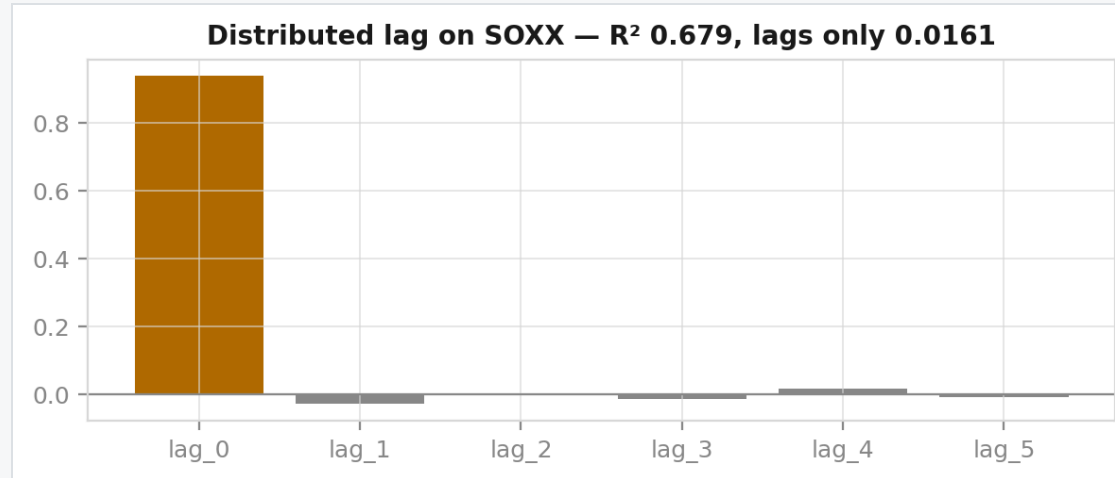


FACTOR	SAME DAY	LEAD (+1)
SOXX	0.8237	-0.1210
TSM	0.6780	-0.0836
^GSPC	0.6728	-0.1466
^VIX	-0.5253	0.0690

Seven factors, 2,915 common trading days. A tall spike at lag 0 and near-flat ground either side.

ASML moves *with* the semiconductor complex, not *after* it. The lag-1 values are **negative** — reversal and bid-ask bounce, not prediction. A true leader would keep its sign.

Quantifying it: one variable carries everything



SPECIFICATION

 R^2

With the same-day term **0.6795**

Lags only – all a forecaster has **0.0161**

A correlation of 0.82 looks like an opportunity, until you notice *when* it exists.

Removing the one variable you cannot know in advance destroys 97.6% of the explanatory power

Granger: reporting the awkward result

Lags 1–5 per factor, Bonferroni threshold $\alpha = 0.01$.

Four factors are significant: SOXX, TSM, ^GSPC, ^TNX. This appears to contradict the section. It does not:

1. **Significance $\neq$ usefulness.** At $n = 2,915$ a test detects effects explaining well under 1% of variance — precisely what the lags-only R^2 of 0.0161 measures directly.
2. **Part 3 already showed** what becomes of small in-sample effects out of sample.

Granger causality is a claim about **forecast improvement**, not about cause. For a chip-equipment maker and a semiconductor index, a **common driver** is by far the likeliest explanation.

What we found

The direction of ASML is not forecastable

Four independent tools, four different labs, one conclusion:

1. ADF and KPSS agree: I(1)
2. The ACF of returns collapses immediately
3. ARIMA's AR and MA roots nearly cancel
4. No model beats naive across 74 origins, DM $p \geq 0.235$

This is weak-form market efficiency (Fama, 1970)

The magnitude of its moves is

GARCH persistence 0.9864, half-life 50.7 days, ARCH fully absorbed.

And we found where our own model fails

Prediction intervals cover 81.5%, not 95% — the constant-variance assumption breaking in plain sight.

The failure mode we deliberately avoided: tuning until something wins. Given enough specifications, one always will.

Limitations & further work

- Single asset, single regime — 2015–2026 was one extraordinary bull run
- Linear models only; non-linear or regime-switching structure would be invisible
- **SARIMA not fitted** — seasonal strength 0.098 did not justify it, and it is the natural next step
- Daily closes hide microstructure; bid-ask bounce likely explains the negative lag-1 values
- Live option chain reflects quotes at one moment; no transaction costs modelled

Reproducibility: every figure and number comes from `make_figures.py`, which imports the same modules that serve the dashboard. One implementation — a slide cannot disagree with the app.

Thank you — questions?

LIVE DASHBOARD · 6 SECTIONS · FASTAPI + REACT

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